

List of Practical		
Sr. No	Program Title	CO Mapping
1.	Implementation of Linux Commands Introduction of Unix/Linux Operating system and their architecture Display system information using uname, hostname, and date etc. File operations using cat, touch, cp, mv, rm, and chmod ,umask etc. Create, view, and navigate directories using mkdir, rmdir, cd, pwd, ls etc. Disk Commands df,du,mount,unmount,mkfs,fsck etc. Use redirection and piping in commands File compression and archiving using tar, gzip, zip, unzip etc. Process commands ps,kill, killall,nice, pgrep, top,htop etc. Network commands ifconfig, ping, netstat, host,ip route etc. Administrator Commands Adduser,Passwd, deluser, usermod, groupadd etc	CO1
2.	Shell Scripting Programming Write a shell script to ask your name, program name and enrollment number and print it on the screen. Write a shell script to find the sum, the average and the product of the four integers entered. write shell script to find average of numbers given at command line Write a shell program to exchange the values of two variables Write a shell program to Print Numbers 1 to 10 using while & do while loop. Write a shell program to Print Numbers 1 to 10 using for loop. Write a shell script to display the digits which are in odd position in a given 5-digit number. Write a shell program to search for a given number from the list of numbers provided using binary search method. Write a shell program to concatenate two strings and find the length of the resultant string	CO1

	Write a shell script to find the smallest of three numbers Write a shell program to count number of words, characters, white spaces and special symbols in a given text	
	Process & Thread Management	
3.	Introduction to C Programming (Statement, Conditional Statement, Loop, Array & Function)	C02
4.	Implement FCFS CPU Scheduling algorithm.	C02
5.	Implement the SJF CPU Scheduling algorithm (For both Pre-emptive and non-pre-emptive).	C02
6.	Implement PRIORITY CPU Scheduling Algorithm (For both Pre-emptive and non-pre-emptive).	C02
7.	Implement Round-Robin CPU Scheduling Algorithm	C02
8.	Implement Multi-Level Queue CPU Scheduling algorithm.	C02
9.	Implement Multilevel Queue CPU Scheduling Algorithm.	C02
	Concurrency and Deadlock Management	
10.	Execute the RACE Condition of Process Synchronization.	C02
11.	Implement the Producer–consumer problem using semaphores.	C02
12.	Design a code and implement the Dining Philosopher problem.	C02
13.	Implement Banker’s algorithm of Deadlock Avoidance.	C02
14.	Execute an algorithm for Deadlock Detection.	C02
	Memory Management	

15.	Implement Contiguous memory variable size partition scheme.	C03
16.	Simulate the First-Fit contiguous memory allocation technique.	C03
17.	Simulate the Best-Fit contiguous memory allocation technique.	C03
18.	Simulate the Worst-Fit contiguous memory allocation technique.	C03
19.	Implement the Non-contiguous Memory Allocation by using Paging.	C03
20.	Implement Contiguous memory variable size partition scheme.	C03
	Page Replacement	
21.	Write a Program to simulate the FIFO page replacement algorithm.	C03
22.	Write a Program to simulate the LRU page replacement Algorithm.	C03
23.	Write a Program to simulate the Optimal page replacement Algorithm.	C03
	Disk Scheduling	C03
24.	Write a program to simulate FCFS Disk Scheduling Algorithm.	C03
25.	Write a Program to simulate the SSTF Disk Scheduling Algorithm.	C03
26.	Write a program to simulate SCAN Disk Scheduling Algorithm.	C03
27.	Write a Program to simulate the C SCAN Disk Scheduling Algorithm.	C03
28.	Write a Program to simulate the LOOK Disk Scheduling Algorithm.	C03
	Modern Operating System	C03

29.	Introduction of CUDA Programming.	C03
30.	Write a program in CUDA print message “Welcome CUDA programming”	C03
31.	Implement matrix multiplication using shared memory in CUDA.	C03
32.	Connects to VMware vCenter and lists all virtual machines along with their power state.	C03
33.	Create a new virtual machine in Azure with specified configurations.	C03
34.	Deploy a simple HTTP-triggered distributed Azure Function.	C03
Total Hours:		48
*Competitive coding list will be shared with the students.		